

1. Bluetooth LE Primer — Bluetooth SIG

<https://www.bluetooth.com/bluetooth-le-primer/>

Use this for the overall BLE architecture, the 40 BLE channels, adaptive frequency hopping, and how BLE shares the 2.4 GHz band with technologies such as Wi-Fi. BLE uses 37 general-purpose/data channels plus advertising channels, and connected devices hop between channels over time.

2. Nordic Bluetooth Low Energy Fundamentals

<https://academy.nordicsemi.com/courses/bluetooth-low-energy-fundamentals/>

This is a structured course covering BLE PHY operation, advertising, connections, packets, and tools such as BLE sniffers. Focus mainly on **Lesson 1: Bluetooth LE Introduction** and **Lesson 2: Bluetooth LE Advertising**.

3. Nordic — Advertising Process

<https://academy.nordicsemi.com/courses/bluetooth-low-energy-fundamentals/lessons/lesson-2-bluetooth-le-advertising/topic/advertising-process/>

Important for understanding how BLE devices announce themselves. BLE advertising packets are sent periodically, and the three primary advertising channels are **37, 38, and 39**.

4. Nordic — Advertisement Packet Structure

<https://academy.nordicsemi.com/courses/bluetooth-low-energy-fundamentals/lessons/lesson-2-bluetooth-le-advertising/topic/advertisement-packet/>

Use this to understand what is actually contained inside a BLE advertising packet, including the header, payload, and packet type.

GitHub to Explore

JiaoXianjun/BTLE

<https://github.com/JiaoXianjun/BTLE>

This is the most relevant repository for the RadioHound direction. It is an open-source BLE SDR project containing BLE baseband algorithms, Python/Verilog implementations, a BLE receiver/sniffer, and a transmitter. It supports the standard **1 Mbps GFSK BLE PHY**, advertising packets, data packets, and channel hopping. The receiver can monitor advertising channels or follow a BLE connection as it hops between frequencies.

What to Look for in the GitHub

Focus on:

- How BLE packets are generated.
- How **GFSK modulation** is implemented.
- How the receiver detects the beginning of a BLE packet.
- How BLE advertising channels are selected.

- How channel hopping is tracked.
- What raw IQ/sample data the receiver needs.
- Which parts could eventually be recreated in Python using RadioHound IQ captures.

First Simulation Goal

The first simulation should be simple:

Generate BLE packet → GFSK modulate → add noise → detect BLE signal → identify packet location